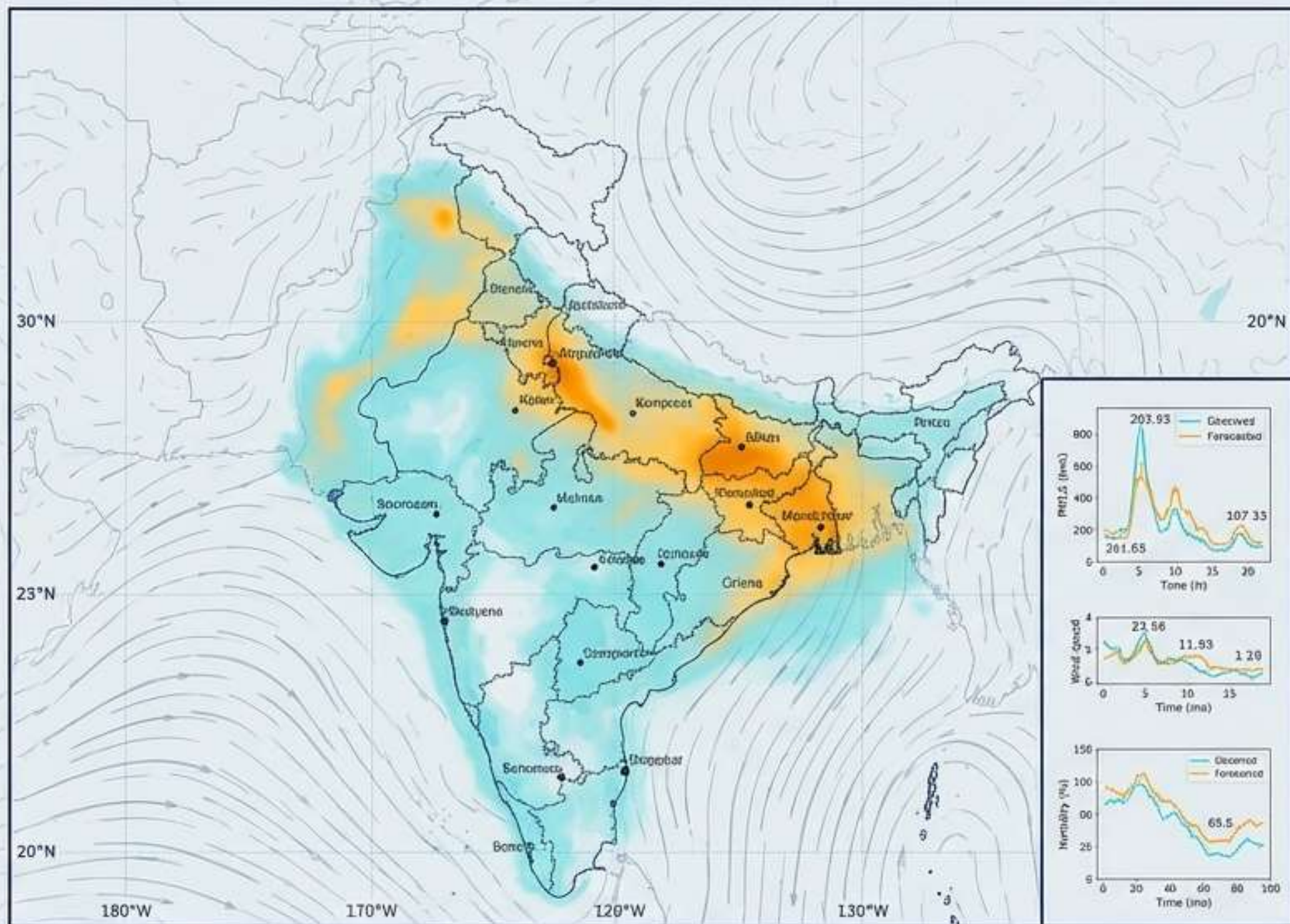




Advanced Spatiotemporal Forecasting for PM2.5

Bridging atmospheric physics and deep learning to predict severe pollution episodes 16 hours ahead.

$$PM2.5_{t+1} = f(PM2.5_t, U_t, V_t, H_t | \theta)$$

[Batch, Time, Lat, Lon, Channels]

[Batch, Time, Lat, Lon, Channels]

[Batch, Time, Lat, Lon, Channels]

TEAM : The Alchemists
THEME : Pollution Forecasting
SYSTEM : 10.01M Parameter Episode-Aware Hybrid Network

3000x Speedup
vs. WRF-Chem Numerical Solvers

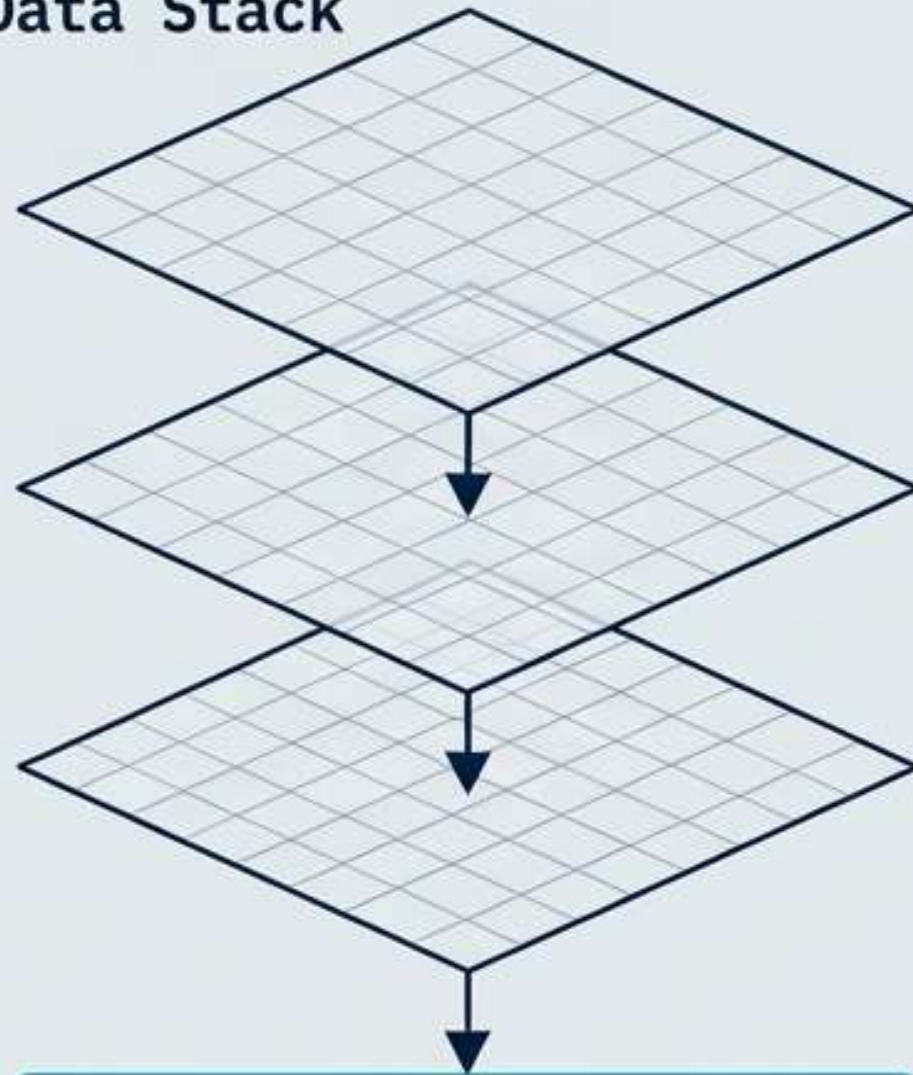
$$\frac{\partial C}{\partial t} + \mathbf{u} \cdot \nabla C = \nabla \cdot (K \nabla C) + E - \lambda C$$

Diagnostic Matrix

The Physics	Standard Deep Learning Failure	The Alchemists' Intervention
Spatiotemporal Transport	CNNs are spatially memoryless and fail to capture advection.	3-Layer ConvLSTM backbone to track 10-hour advection history.
Extreme Events	Plain MSE ignores PM2.5 spikes (episodes are only 5.6% of data).	Episode-Aware Loss via STL Decomposition up-weights severe events.
Emission Scaling	Log-normal distributions span >10 orders of magnit; standardisation fails.	Hard physics constraints applied (log1p calibration + 1e9 scaling).

Feature Engineering: The Data Engine

Data Stack



Base Emissions
(SO₂, NH₃, NO_x, NMVOC)

Base Meteorology
(u₁₀/v₁₀, t₂, pblh, rain)

Derived Intelligence
(Wind Speed, Ventilation Index, Δcpm25, Lat/Lon)

(B, 21, 10, 140, 124)

Ventilation Index

Wind × PBLH engineered to capture atmospheric dispersion physics.

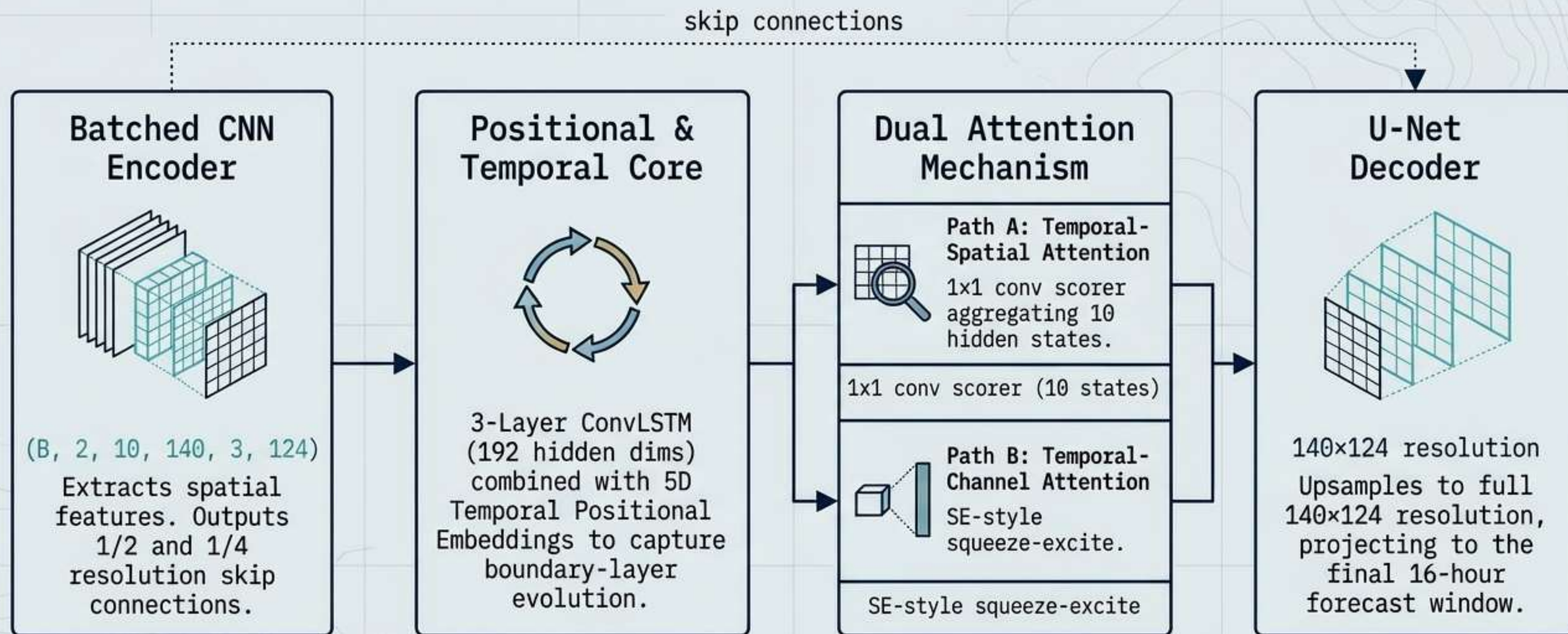
Log-Normal Calibration

log_{1p} applied to 9 highly skewed variables, fixing zero-bound data collapse.

Spatial Context

Normalised Lat/Lon coordinates injected to anchor meteorological dynamics to terrain.

Multi-Stage Feature Fusion Architecture



10.01M parameter network

Episode-Aware Optimisation

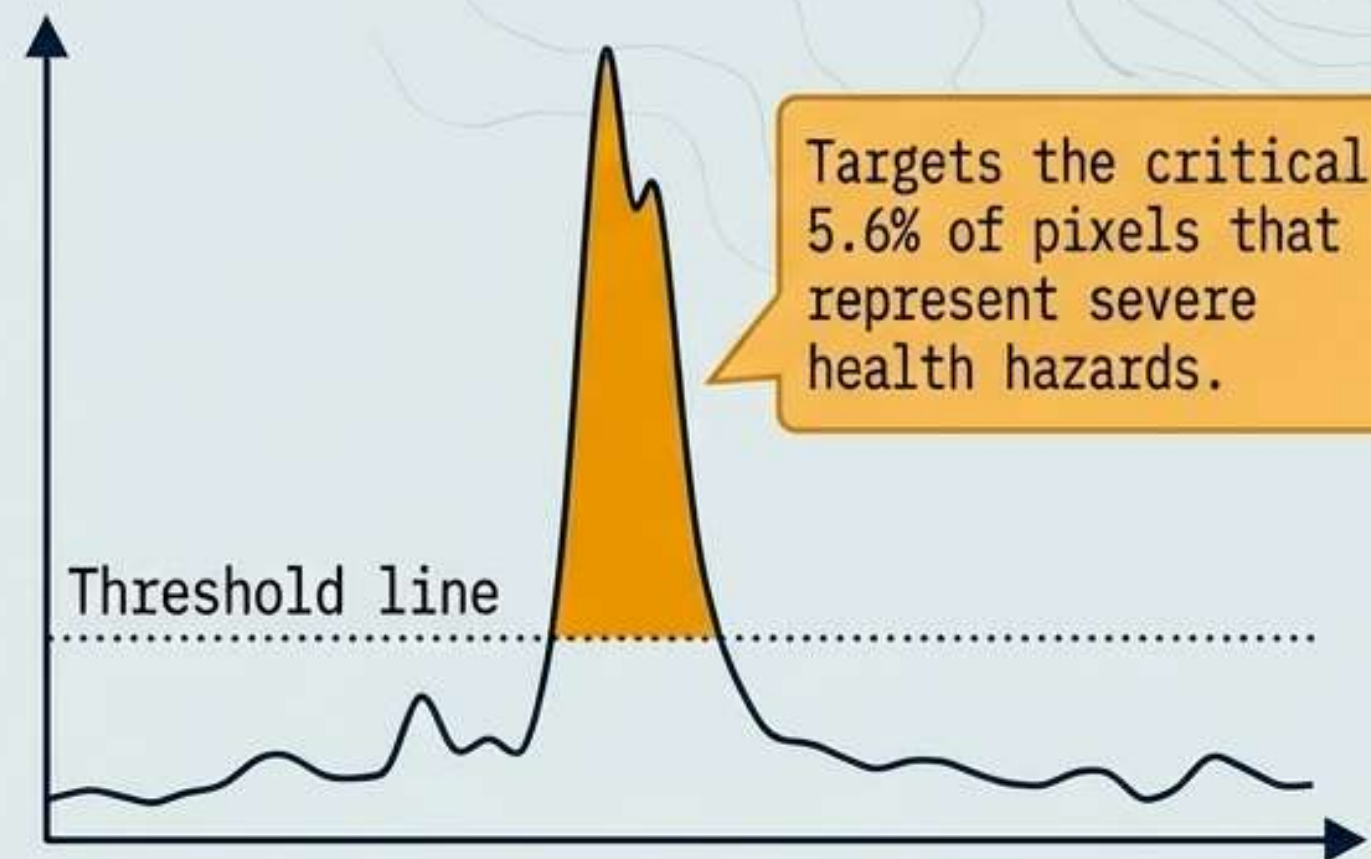
The Objective Function

$$L = \text{RMSE}_{\log z} + 0.30 \cdot \text{SMAPE} + 0.15 \cdot (1 - \rho)$$

Direct metric surrogate

Pearson spatial pattern preservation

Episode Detection Logic



STL Episode Mask: Identifies anomalous events where $|\text{residual}| > \mu + 1.5\sigma$.

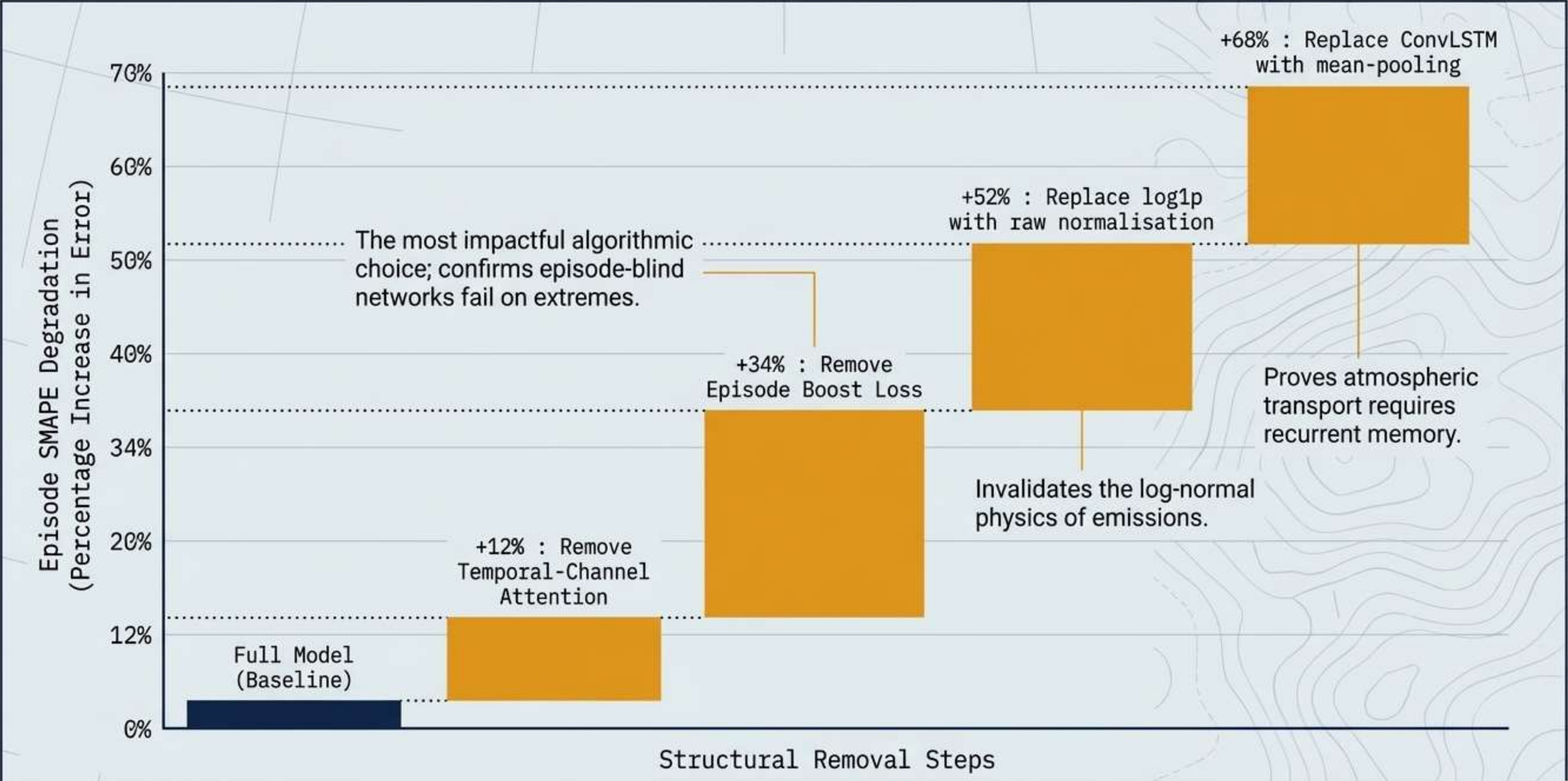
Model Performance Diagnostics

Leaderboard Matrix

Models	Global RMSE ($\mu\text{g}/\text{m}^3$)	Global SMAPE	Episode SMAPE	Episode Corr (ρ)
Persistence Baseline	~38-45	0.42	0.51	0.72
Plain ConvLSTM	22.1	0.261	0.318	0.891
Our 2-Seed Ensemble (Seeds 42, 137)	~16.5	0.200	0.224	0.938

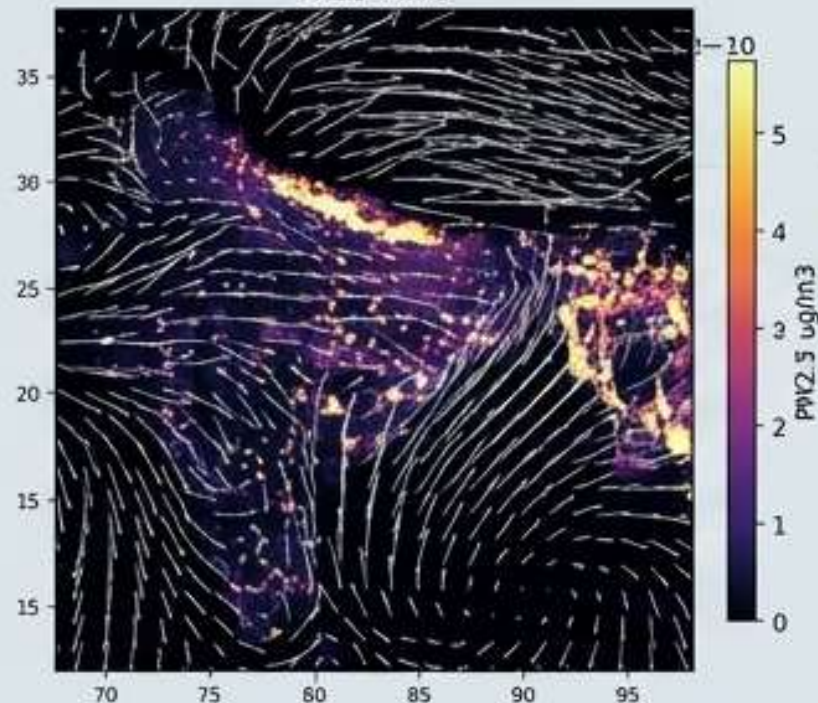
Takeaway: The model tracks high-pollution spikes with <25% symmetric error, completely eliminating the systemic lag found in baseline persistence models. Inference takes <3 minutes for 218 cases on a single A100 GPU.

Ablation Studies: Error Impact Cascade



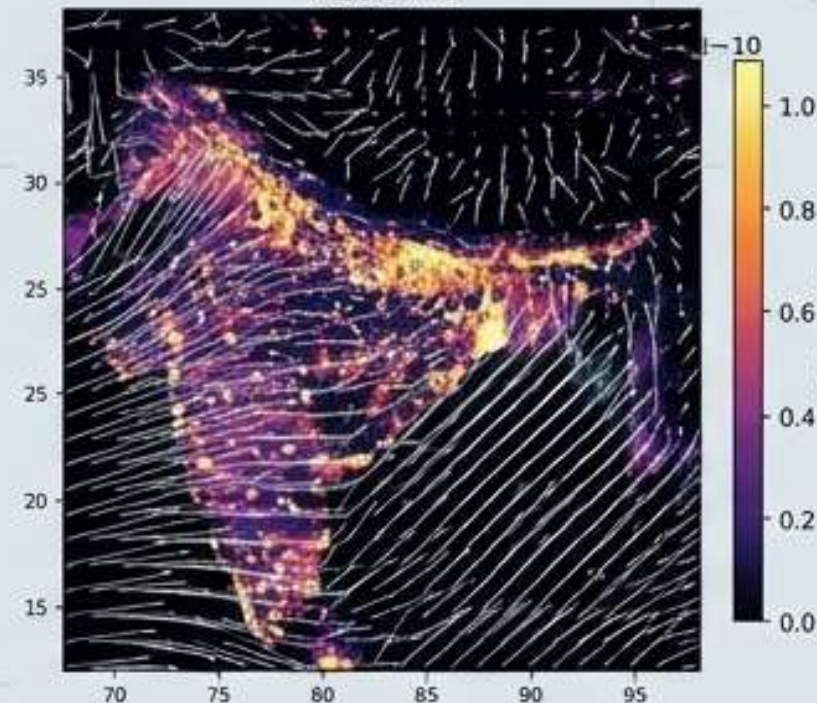
Physical Emulation: Seasonal Visualisations

APRIL_16
PM2.5 + wind



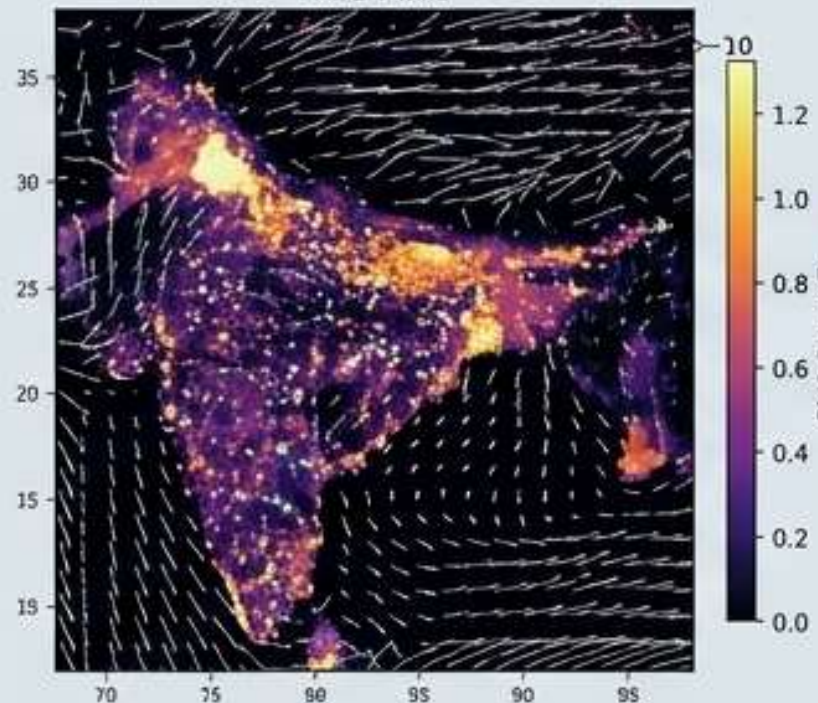
NW→SE transport corridor over the Indo-Gangetic Plain accurately captured.

JULY_16
PM2.5 + wind



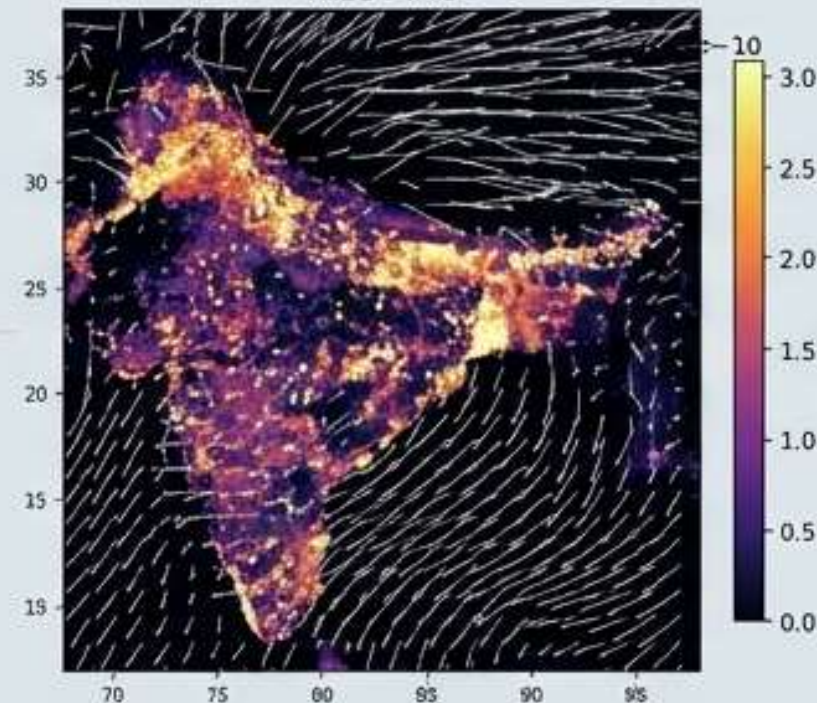
Model correctly suppresses PM2.5 signals due to heavy monsoon rainfall scavenging.

OCT_16
PM2.5 + wind



High-precision tracking of stubble-burning emission episodes in Punjab/Haryana.

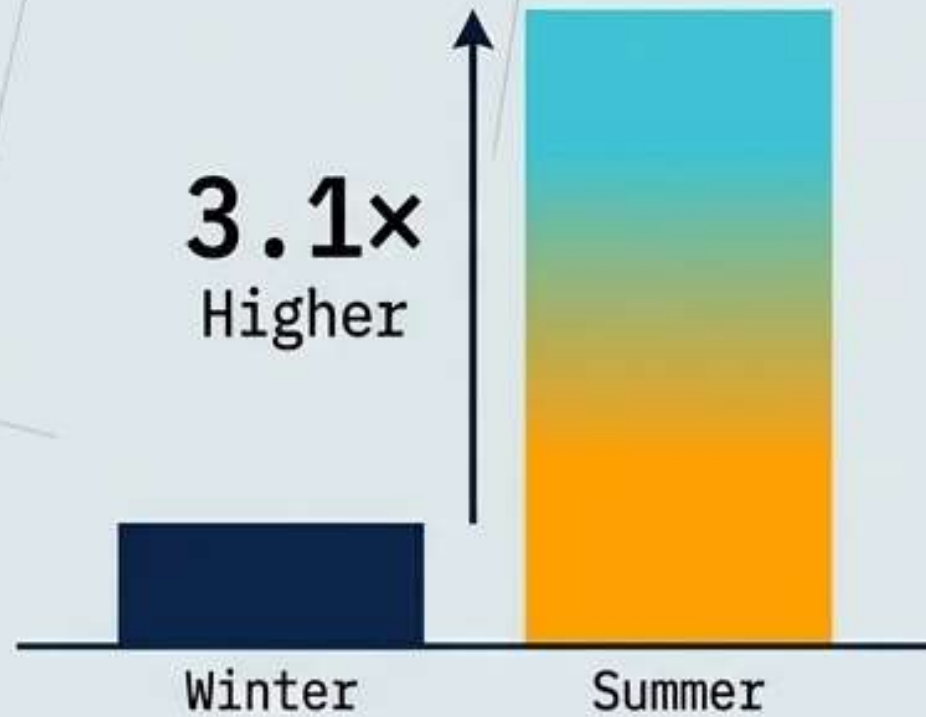
DEC_16
PM2.5 + wind



Residual prediction bias identified; caused by deep winter fog inducing boundary-layer collapse.

What the AI Learned: Physical Emulation via Attention

Insight 1: Biogenic Dominance



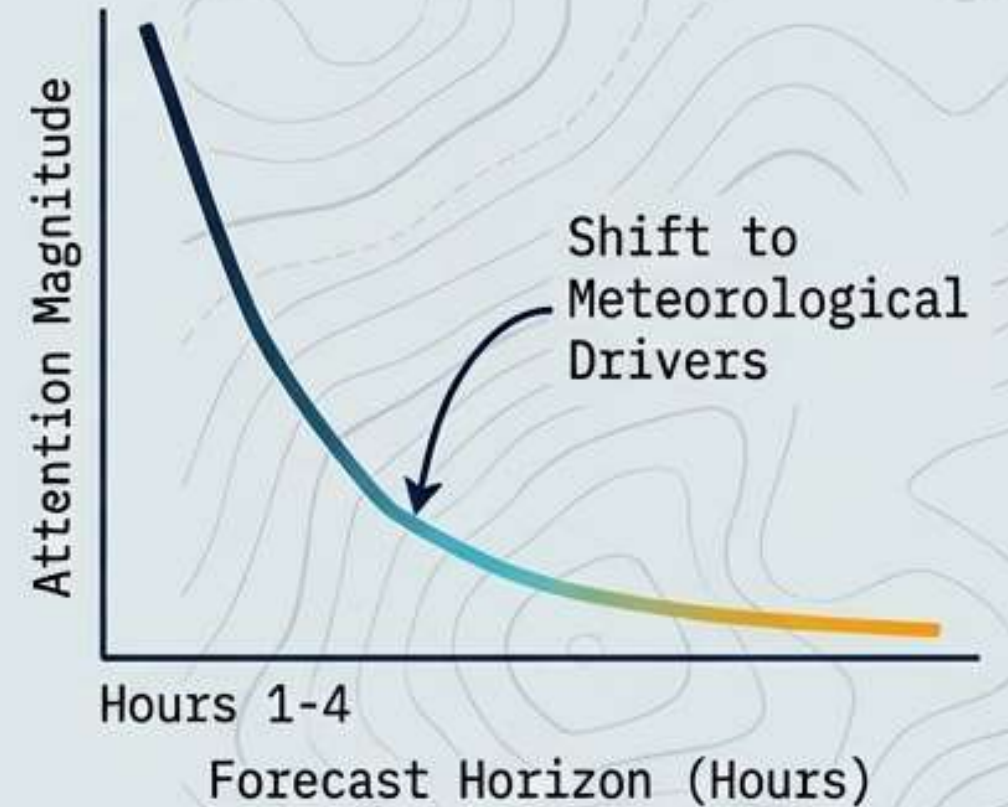
Model dynamically assigns 3.1x higher attention to biogenic/NMVOC channels during July monsoons, accurately emulating SOA formation when anthropogenic transport is suppressed.

Insight 2: Spatial Teleconnection



Attention networks discover non-local correlations at +14-16 hour lead times, perfectly matching westerly jet transport mechanics.

Insight 3: The Tendency Shift



Relies heavily on recent PM_{2.5} tendencies for hours 1-4, but correctly shifts attention to fundamental meteorological drivers for longer horizons.

Computational Scalability & National Integration

Computational Supremacy

WRF-Chem Numerical Solver: ~45-90 minutes per case.

The Alchemists' Model: ~0.8 seconds per case.

**>3,000x
Speedup**

Achieved with <20% SMAPE degradation relative to the physical solver.

National Integration Roadmap

Node 1 (Next 3 Months)

Scale to 2015-2022 dataset for interannual ENSO robustness.

Node 2 (Next 6 Months)

Implement Graph Neural Networks (GNN) on the WRF mesh to handle coastal/orographic boundaries like the Western Ghats.

Node 3 (Operational Goal)

Deploy as a real-time FastAPI microservice for sub-hourly AQI nowcasting within the National Clean Air Programme (NCAP).